

OREN SASON

Durham, United Kingdom | U.S. Citizen

+44 7584 488005 | orensason.u@gmail.com

Engineering Portfolio: <https://orensasonu-eng.github.io/>

Mechanical Engineering student at Durham University with a background in systems-based design, technical analysis and intelligence-style analytical research, with a strong interest in defence, security, and high-reliability engineering applications.

EDUCATION

Durham University, UK | October 2024 - June 2027 Expected

BEng Mechanical Engineering

Modules include: Solid Mechanics and Systems, Thermodynamics and Fluid Mechanics, Electrical Systems, Electronics, Engineering Mathematics and Engineering Practice

- Built a strong understanding of engineering systems, including mechanical, thermal, and electrical behaviour
- Applied engineering mathematics to modelling, analysis, and system-level problem-solving
- Developed understanding of system dynamics, energy transfer, and structural response in engineering applications

Columbia Business School — Venture for All | June 2023

- Entrepreneurship and Innovation Program
- Worked on developing new start-up concepts
- Assessed feasibility of real-world constraints in early-stage projects
- Worked in teams to put together project concepts and deliver formal presentations

King's College London — STEM Summer School | July 2022

- Studied Digital Logic, Robotics, AI Fundamentals, Microprocessors and Medical 3D Design
 - Applied concepts in system design, basic programming and prototyping
-

TECHNICAL SKILLS

Programming & Modeling: C++, MATLAB (numerical methods & simulation)

Engineering Tools: LTspice, CAD

Systems & Analysis: Circuit Simulation, Nonlinear System Modeling, Technical Analysis and Report Writing

Research & Intelligence Analysis: Open-source intelligence and geopolitical research reporting

PROJECTS

Portable Insulin Cooler (Group Design Project) | 2025

- Designed a portable medical refrigeration unit to store insulin at safe temperatures (2–8°C) for up to 40 hours without continuous power
- Applied a full systems engineering design approach, translating real-world constraints into requirements and using trade-off analysis to determine the final concept
- Integrated thermal, electrical, and structural subsystems
- Developed a hybrid cooling system using phase change materials and thermoelectric module (Peltier device) for operation in harsh environments
- Produced engineering documentation including bill of materials, cost analysis, manufacturing considerations, user journey analysis and a case study
- Validated system performance through thermal modelling and structural testing, ensuring suitability for remote and resource-constrained environments

Chua Chaotic Oscillator Simulation (LTspice) | 2025

- Modeled and simulated a chaotic nonlinear electronic systems using Ltspice
 - Analyzed unpredictable system behavior and instability with MATLAB under varying parameters
 - Investigated system stability under varying conditions
-

EXPERIENCE

Research Analyst — Durham Diplomatic Society | October 2025 - Present

- Conduct research on geopolitical and security developments across Europe and the Middle East
- Produce analytical summaries for weekly publication to an audience of approximately 3,000 readers
- Collaborate within a 40-member analytical team
- Participate in discussions with senior diplomatic and intelligence figures including former intelligence officers and ambassadors
- Deliver presentations and contribute to policy-focused analytical discussions

Engineering Intern — CAVO Automotive | July 2023 - August 2023

- Supported the manufacture and quality assurance of precision engineered components for premium OEM clients including major automotive brands
 - Assisted with assembly, inspection, and measurement processes to verify compliance with strict tolerances and quality standards
 - Contributed to technical documentation and gained hands-on experience in industrial manufacturing workflows
-

TRAINING & CERTIFICATIONS

- Counterintelligence Awareness Training — U.S. Department of Defense (2025)
 - Counterintelligence Reporting Course — U.S. Department of Defense (2025)
 - Nuclear Security Threats & Risks — IAEA (2025)
 - IAEA Safeguards Fundamentals — IAEA (2025)
 - OSINT Foundations — ALCON Intelligence (2025)
-

PROFESSIONAL AFFILIATIONS

- Royal Aeronautical Society
 - Institution of Engineering and Technology
 - Durham Union
-

LANGUAGES

English (Fluent) | Turkish (Fluent) | Spanish (Basic)